

Linear Span (or) Linear hull

Let S be a non-empty subset of vector space $V(F)$.

The linear span of S is defined as the set consisting of all linear combinations of finite subset of the elements of S and is denoted by $L(S)$

Symbolically,

$$L(S) = \left\{ u \mid u = \sum a_i u_i, u_i \in S, a_i \in F \right\}$$

Here $L(S)$ is also called the set spanned or generated by S

Some Important theorems

1. The linear span $L(S)$ of any non-empty subset S of vector space $V(F)$ is a subspace of $V(F)$.

2. Let M and N be subspaces of a vector space $V(F)$. Then

(i) $M \subset M+N$

(ii) $N \subset M+N$

(iii) $M+N = L(M \cup N)$

3. If M and N are non-empty subset of a vector space $V(F)$, then

(i) $M \subset N \Rightarrow L(M) \subset L(N)$

(ii) $L(M \cup N) = L(M) + L(N)$

(iii) M is a subspace of V

$$\Leftrightarrow L(M) = M$$

(iv) $L\{L(M)\} = L(M)$
 $\quad \quad \quad = x =$

1. Determine the subspace of R^3 spanned by the vectors $\alpha = (1, 2, 3)$, $\beta = (-1, 2, 4)$ show that $\gamma = (3, 2, 2)$ is in the subspace

solution:

$$\text{let } S = \{ \alpha, \beta \} \\ = \{ (1, 2, 3), (-1, 2, 4) \}$$

$$L(S) = L\{ \alpha, \beta \} \\ = \{ c\alpha + d\beta \mid c, d \in \mathbb{R} \}$$

$$c\alpha + d\beta = c(1, 2, 3) + d(-1, 2, 4)$$

If $\gamma \in L\{ \alpha, \beta \}$ then there must be real numbers c, d such that

$$\begin{aligned} (3, 2, 2) &= c\alpha + d\beta \\ &= c(1, 2, 3) + d(-1, 2, 4) \\ &= (c-d, 2c+2d, 3c+4d) \end{aligned}$$

$$\text{Now } c-d = 3 \quad \text{--- (1)}$$

$$2c+2d = 2 \quad \text{--- (2)}$$

$$3c+4d = 2 \quad \text{--- (3)}$$

these equations are consistent and their solution is $c = 2, d = -1$

Hence $\gamma \in L\{ \alpha, \beta \}$

i.e. γ belongs to the subspace generated by α, β

2. Find the condition among x, y, z such that the vector (x, y, z) belongs to the space generated by $\alpha = (2, 1, 0)$, $\beta = (1, -1, 2)$, $\gamma = (0, 3, -4)$

solution:

If $(x, y, z) \in L\{\alpha, \beta, \gamma\}$,

then (x, y, z) can be expressed as a linear combination of α, β, γ

$$\text{Let } (x, y, z) = c_1 \alpha + c_2 \beta + c_3 \gamma$$

$$\text{then } (x, y, z) = c_1 (2, 1, 0) + c_2 (1, -1, 2) + c_3 (0, 3, -4)$$

$$(x, y, z) = (2c_1 + c_2, c_1 - c_2 + 3c_3, 2c_2 - 4c_3)$$

This gives,

$$2c_1 + c_2 = x \quad \text{--- (1)}$$

$$c_1 - c_2 + 3c_3 = y \quad \text{--- (2)}$$

$$2c_2 - 4c_3 = z \quad \text{--- (3)}$$

Multiplying second equation by 2 and subtracting from equation (1)

$$\begin{aligned}
 & 2c_1 + c_2 = x \\
 (-) \quad & \underline{2c_1 - 2c_2 + 6c_3 = 2y} \\
 & 3c_2 - 6c_3 = x - 2y \\
 \Rightarrow \quad & c_2 - 2c_3 = \frac{x - 2y}{3} \quad \text{--- (4)}
 \end{aligned}$$

from (3) $2c_2 - 4c_3 = z$

$$c_2 - 2c_3 = \frac{z}{2} \quad \text{--- (5)}$$

$$\therefore \frac{x - 2y}{3} = \frac{z}{2}$$

$$\Rightarrow 2x - 4y = 3z$$

$$\Rightarrow 2x - 4y - 3z = 0$$

which is required condition

$$\underline{= x =}$$

3. Determine the subspace of $\mathbb{R}^3$ spanned by the vectors $\alpha = (1, 2, 3)$, $\beta = (3, 1, 0)$.
Examine, if $\gamma = (2, 1, 3)$ and $\delta = (-1, 3, 6)$ are in the subspace.

Solution:

Let $S = \{ \alpha, \beta \}$, where $\alpha = (1, 2, 3)$
 $\beta = (3, 1, 0)$

$$L(S) = L\{\alpha, \beta\}$$

$$= \{c\alpha + d\beta \mid c, d \in \mathbb{R}\}$$

$$= \{c(1, 2, 3) + d(3, 1, 0) \mid c, d \in \mathbb{R}\} \subseteq \mathbb{R}^3$$

Let $\gamma \in L(S)$,

then there must be real numbers c, d such that

$$\gamma = c\alpha + d\beta$$

$$(2, 1, 3) = (c + 3d, 2c + d, 3c)$$

Here

$$c + 3d = 2 \quad 2c + d = 1, \quad 3c = 3$$

$$d = -1 \quad c = 1$$

But these two values not satisfy the first equation

so These equations are inconsistent

so γ is not in $L(S)$

Let $\delta \in L(S)$,

then there must be real numbers c, d such that

$$\delta = c\alpha + d\beta$$

$$(-1, 3, 6) = (c + 3d, 2c + d, 3c)$$

Here

$$\begin{array}{rcl} c+3d=-1, & 2c+d=3, & 3c=6 \\ \text{---} \textcircled{1} & \text{---} \textcircled{2} & \text{---} \textcircled{3} \\ & d=-1 & c=2 \end{array}$$

these two values are satisfied equation $\textcircled{1}$

so these equations are consistent

$$\text{so } (-1, 3, 6) = 2(1, 2, 3) - 1(3, 1, 0)$$

$$\therefore \gamma = 2\alpha - \beta$$

$$\therefore \gamma \in L(S).$$

= x =

$$4. \text{ If } \alpha = (1, 2, -1), \beta = (2, -3, 2), \\ \gamma = (4, 1, 3), \delta = (-3, 1, 2) \in \mathbb{R}^3(\mathbb{R})$$

Prove that $L\{\alpha, \beta\} \neq L\{\gamma, \delta\}$

solution:

Let, if possible, $L\{\alpha, \beta\} = L\{\gamma, \delta\}$

then $\exists$ scalars $x, y \in \mathbb{R}$ and for arbitrary $a, b \in \mathbb{R}$, such that

$$x\alpha + y\beta = a\gamma + b\delta$$

$$x(1, 2, -1) + y(2, -3, 2) = a(4, 1, 3) + b(-3, 1, 2)$$

$$\Rightarrow (x+2y, 2x-3y, -x+2y) \\ = (4a-3b, a+b, 3a+2b)$$

Here

$$x+2y = 4a-3b \quad \text{---(1)}$$

$$2x-3y = a+b \quad \text{---(2)}$$

$$-x+2y = 3a+2b \quad \text{---(3)}$$

From (1) & (3)

$$\begin{array}{r} x+2y = 4a-3b \\ (-) \quad -x+2y = 3a+2b \\ \hline 2x = a-5b \end{array}$$

$$x = \frac{1}{2}(a-5b)$$

then $y = \frac{1}{4}(7a-b)$

Now

$$2x-3y = 2\left(\frac{1}{2}(a-5b)\right) - 3\left(\frac{1}{4}(7a-b)\right)$$

$$= a-5b - \frac{3}{4}(7a-b)$$

$$= \frac{4a-20b-21a+3b}{4}$$

$$= \frac{-17a-17b}{4}$$

$$= -\frac{17}{4}(a+b) \neq a+b.$$

$\therefore$ these equations are inconsistent

$$\text{so } \vdash \{ \alpha, \beta \} \neq \vdash \{ \gamma, \delta \}$$

$$= x =$$

5. In the vector space $V_3(\mathbb{R})$, consider the vectors $\alpha = (1, 2, 1)$,

$$\beta = (3, 1, 5),$$

$$\gamma = (3, -4, 7).$$

Show that the subspace spanned by $S = \{ \alpha, \beta \}$ and $T = \{ \alpha, \beta, \gamma \}$ are the same.

solution:

$$\therefore S \subseteq T \Rightarrow \vdash(S) \subseteq \vdash(T)$$

Now, To show that γ can be expressed as a linear combination of α, β .

$$\text{let } \gamma = a\alpha + b\beta, \text{ for some scalars } a, b \in F.$$

$$\text{then } (3, -4, 7) = a(1, 2, 1) + b(3, 1, 5)$$

$$(3, -4, 7) = (a+3b, 2a+b, a+5b)$$

$$\Rightarrow a+3b=3 \quad \text{---(1)}$$

$$2a+b=-4 \quad \text{---(2)}$$

$$a+5b=7 \quad \text{---(3)}$$

$$\Rightarrow a = -3, b = 2$$

$$\therefore \gamma = -3\alpha + 2\beta.$$

Now, δ be an arbitrary element of $L(T)$

$$\text{then } \delta = a_1\alpha + a_2\beta + a_3\gamma$$

for some $a_1, a_2, a_3 \in \mathbb{F}$.

$$= a_1\alpha + a_2\beta + a_3(-3\alpha + 2\beta)$$

$$= (a_1 - 3a_3)\alpha + (a_2 + 2a_3)\beta \in L(S)$$

$$\delta \in L(T) \Rightarrow \delta \in L(S)$$

$$L(T) \subseteq L(S)$$

$$\therefore L(S) = L(T).$$

So the subspace spanned by

$S = \{\alpha, \beta\}$ and $T = \{\alpha, \beta, \gamma\}$ are
same

 $\times$

6. Let $S = \{\alpha, \beta, \gamma\}$ and $T = \{\alpha, \beta, \alpha + \beta, \beta + \gamma\}$
be subsets of a real vector space V .

